

UNIVERSIDAD DEL PACÍFICO

Voluntary Liquidity Buffers and Public Dollar Backing

Carlos Gómez Puicán

AI & Economic Modeling

Joint IE1 baseline with Fabián Méndez Suárez; supervised by Marco Ortiz.

Optimization closure and new derivations: Carlos's individual project.

Does public backing change self-insurance?

How does public dollar-liquidity backing affect a bank's voluntary liquidity buffer, conditional on interbank conditions?

Broader joint IE1 question

Who contributes to and uses liquidity coverage when banks differ in risk and access?

Individual project's first step

Determine the bank's self-insurance choice before studying aggregate feedbacks.

Banks differ in exposure and market access

Banks and currencies

- Two bank types: L and S
- Two currencies: soles and dollars
- Different liquidity-shock exposure
- Different interbank access

Liquidity held before the shock

ρ Required reserves

u Voluntary buffer

Deposits, currency shares, and equity are fixed in the extension.

Dollars face normal and sudden-stop aggregate states; soles have no aggregate shock in this baseline.

Liquidity protection has three layers

Voluntary buffer



Liquidity shock



Interbank market



Public backing



Uncovered deficit

Own self-insurance

Liquidity held before the shock

External liquidity insurance

Interbank funding, then public coverage

Any remaining deficit carries a private loss.

The extension makes the buffer a private choice

Inherited: joint IE1 framework

- Balance sheets and reserves
- Liquidity shocks
- Interbank matching
- Public coverage

New: Carlos's individual project

- Private expected-cost objective
- Opportunity cost of u
- Costs of funded and uncovered deficits
- Optimal buffer and conditional result

The joint framework specified allocations; it did not yet specify the bank objective determining u .

Choose liquidity to minimize private cost

$$\min_{0 \leq u_k^c \leq \bar{u}_k^c} \kappa_k^c u_k^c + \mathbb{E}_A[h_k^c(A) L_k^c(u_k^c, A)]$$

κ_k^c : net opportunity cost L_k^c : expected deficit per deposit h_k^c : cost per unit of deficit

For a given bank and currency, deficit costs split into:

$$h(A) = \underbrace{\Psi q_I}_{\text{interbank-funded}} + \underbrace{(1 - \Psi)\varphi q_F}_{\text{publicly funded}} + \underbrace{(1 - \Psi)(1 - \varphi)\ell}_{\text{uncovered}}.$$

Ψ : interbank fraction φ : public fraction of the residual deficit.

Atomistic bank; fixed balance-sheet inputs; no surplus income.

Nonnegative credit: $\bar{u}_k^c = 1 + E_k^c/d_k^c - \rho^c \geq 0$, for $d_k^c > 0$.

Self-insurance balances cost and avoided deficits

Extra liquidity reduces deficits when the bank is short.

$$\frac{\partial L(u, A)}{\partial u} = -\pi^-(u, A)$$

At an interior optimum:

$$\kappa = \mathbb{E}_A[h(A)\pi^-(u^*, A)]$$

Marginal opportunity cost of self-insurance
= expected marginal deficit cost avoided

π^- : conditional deficit probability. One bank and currency; continuous shocks.

Public backing can reduce self-insurance

Hold aggregate P , Q , and Δ fixed across backing scenarios.

$$\frac{du_k^{u,*}}{dR_D} < 0$$

More effective public coverage
reduces the cost of being short.

- Regular interior optimum: positive curvature, away from kinks
- Uncovered needs cost more: $\ell^u > q_F^u$
- In one positive-probability state:
 $\pi_k^{u-} > 0, \quad \Psi_k^{-,u} < 1, \quad 0 < F^u < \Delta^u$
- Rationed public coverage responds:
 $\partial \varphi^u / \partial R_D = 1 / \Delta^u > 0$

Also fixed: policy, shocks, costs, deposits, and feasible bounds.

Conditional individual best response; no equilibrium sign theorem.

Required reserves protect banks or the collective

NU: non-utilizable

Own usable liquidity

u

Required reserves contribute to collective dollar backing.

$$F^u = \nu_F^u \rho^u D^u + R_D$$

U: utilizable

Own usable liquidity

$u + \rho$

Required reserves do not also enter the collective fund.

$$F^u = R_D$$

Direct individual protection versus collective protection

The same reserve principal is counted once. No unconditional ranking of optimal buffers.

The final project must close the system response

- Discipline private cost primitives: κ , q_I , q_F , ℓ
- Solve the aggregate buffer fixed point
- Let P , Q , Δ , and coverage respond
- Study utilization across bank types and currencies
- Discuss net redistribution only with complete contractual accounting

The conditional self-insurance mechanism is derived under the baseline assumptions.

The system-wide response remains the next modeling task.